

CAE-YOLOV8: Occlusion Object Detection based on improved YOLOv8

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Abstract—Occlusion recognition plays a vital role in smart manufacturing and industrial automation. This paper proposes an enhanced YOLOv8-based model designed to accurately recognize and locate occluded objects in industrial environments, assisting robots in precise grasping tasks. The model integrates the AKConv module and the Effciou loss function, which improve feature extraction and bounding box regression, especially in complex and highly occluded scenarios. Experimental results confirm that the modified YOLOv8s Effciou model achieves a balanced performance across different levels of occlusion, showing notable robustness in challenging environments with high occlusion.

Keywords—component; YOLOv8, Occlusion object recognition, CBAM attention mechanisms, AKConv module, Effciou loss function

I. INTRODUCTION

In recent years, deep learning has made significant progress in object detection. However, in complex industrial environments where objects to be grasped are stacked and occluded, the detection and precise positioning of these objects remain challenging. Several studies have specifically focused on improving object grasping under occlusion. For instance, the TARGO dataset evaluates how occlusion impacts grasping performance, revealing a significant drop in success rates when objects are partially or fully obscured[1]. Another approach applied deep reinforcement learning and multi-view observation techniques, achieving a high grasp success rate of 97.8% in occluded conditions[2]. Additionally, some research combines 3D object detection with occlusion prediction algorithms to boost grasping efficiency in cluttered scenes[3]. While these methods have made notable advancements in grasping tasks, they often face challenges such as real-time performance limitations and a decline in accuracy when handling diverse and rapidly changing industrial conditions. To

address these issues, it became necessary to shift towards more efficient deep learning models that could better handle real-time detection and positioning. YOLO models, known for their fast inference speed and high detection accuracy, present a viable solution for object recognition under occlusion. The integration of attention mechanisms like CBAM and AKConv variable convolution and optimized loss functions such as Effciou further enhances the capability of YOLO models to manage occlusion. Hence, this paper proposes an improved method based on YOLOv8 to enhance object recognition performance specifically in occluded scenes, bridging the gap between traditional grasping methods and more advanced deep learning techniques.

II. CAE-YOLOV8

The CAE-YOLOv8 model introduced in this paper integrates three key enhancements—CBAM (Convolutional Block Attention Module), AKConv (Arbitrary Kernel Convolution), and EffCiou loss function—aiming to improve detection performance under challenging conditions such as occlusion. The model architecture is illustrated in Fig. 1. Compared to the original YOLOv8, CAE-YOLOv8 makes the following improvements: First, the **CBAM attention mechanism** is added after the SPPF layer to **enhance the feature extraction process** by **applying both channel and spatial attention**. Next, the AKConv module is incorporated after the CBAM layer to dynamically adjust the convolution kernel, improving the model's adaptability to objects with varying shapes and sizes. Lastly, the EffCiou loss function replaces the traditional IoU loss, optimizing bounding box regression to achieve higher localization accuracy in complex scenarios. These enhancements collectively strengthen the model's ability to detect and classify objects accurately, even in environments with heavy occlusions.

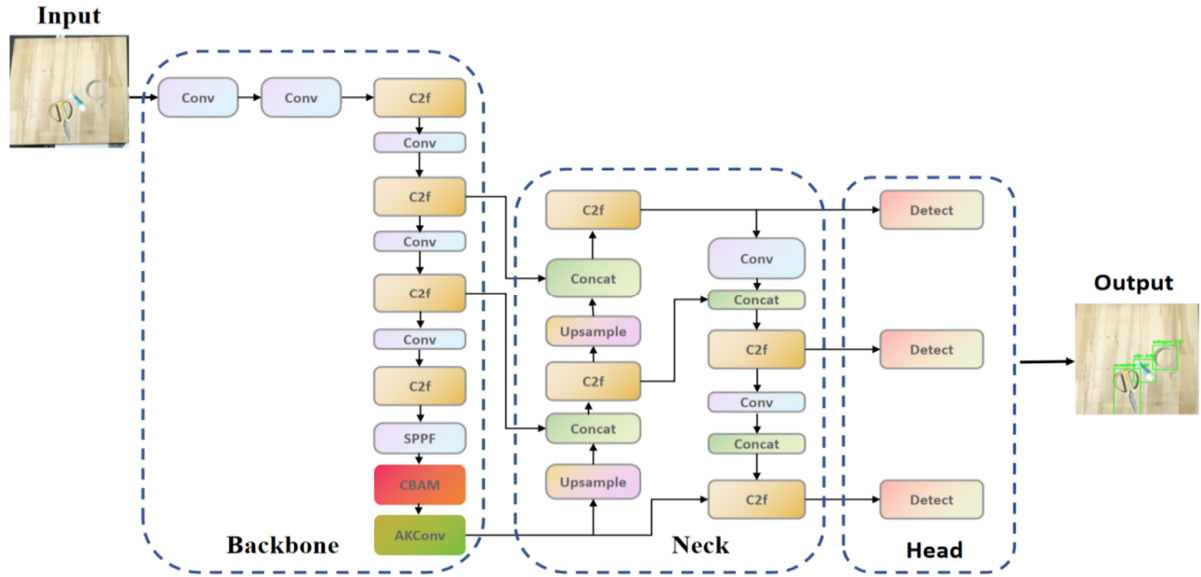


Figure 1. CAE-YOLOv8 model structure

A. CBAM (Convolutional Block Attention Module)

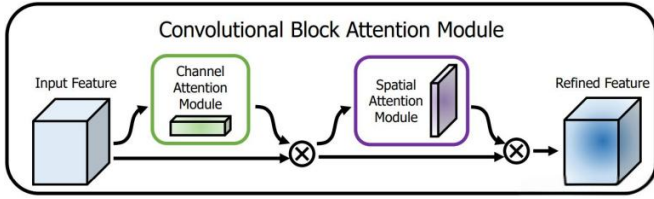


Figure 2. CBAM Attention Mechanism Structure

The CBAM[4] attention mechanism is introduced after the SPPF (Spatial Pyramid Pooling - Fast) layer to improve feature extraction, as shown in Figure 2. It refines the multi-scale features from the SPPF layer by applying channel and spatial attention to feature maps. The channel attention module extracts global information using average and max pooling, assigning attention weights to important channels. The spatial attention module uses convolution to focus on key spatial regions. Figure 2 illustrates how CBAM integrates into the model, highlighting its role in enhancing feature quality through combined channel and spatial attention.

B. AKConv variable convolution

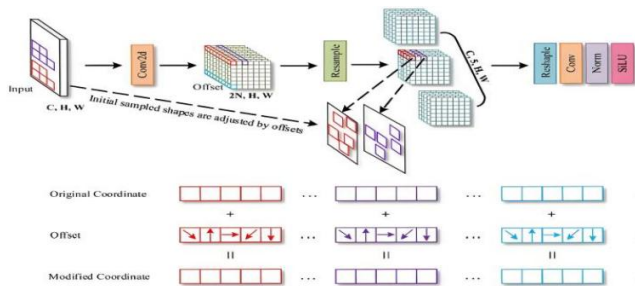


Figure 3. AKConv Variable Convolution

The AKConv[5] variable convolution is introduced after the SPPF layer to boost feature extraction, as shown in Figure 3. It allows dynamic adjustment of sampling positions and shapes based on input data characteristics, making the model more adaptable to varying object sizes and shapes. This adjustment is particularly useful in complex scenarios with multiple targets or occlusions. Figure 3 illustrates how AKConv integrates into the model, demonstrating its role in refining multi-scale feature extraction by adjusting the sampling grid with calculated offsets, thus enhancing adaptability in diverse visual environments.

C. Effciou Loss improvement

The Effciou loss function[6] was used to improve bounding box regression in place of traditional IoU-based losses. Effciou not only considers the overlap between the predicted and ground truth bounding boxes but also introduces a penalty for the distance between their center points. This adjustment helps the model achieve better object localization, particularly in challenging scenarios such as occlusions or complex backgrounds. As shown in the Figure 4, Effciou (ECIOU_Loss) achieves faster convergence compared to CIOU_Loss and EIOU_Loss, leading to lower loss values throughout training. This indicates more precise adjustments to the bounding box positions during the learning process, resulting in improved detection accuracy. By integrating distance, size, and position considerations into the loss calculation, Effciou refines the model's ability to handle nuanced variations in object positions, ensuring better detection under varying conditions. The formula is as follows:

$$\text{Effciou}(B, B_{gt}) = 1 - \text{IoU}(B, B_{gt}) + \frac{d^2(B, B_{gt})}{c^2}$$

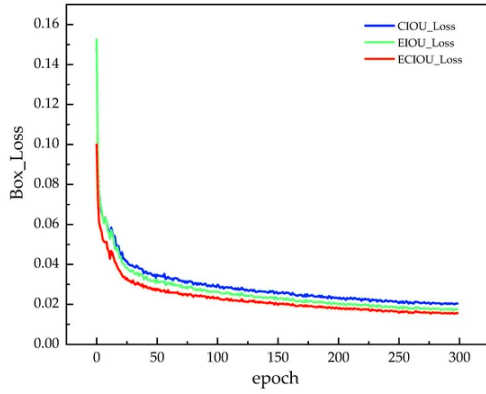


Figure 4. Comparison of loss function convergence

D. Custom checking codes with different occlusion levels

We used a custom method to evaluate the model's accuracy under varying occlusion levels. The model generates predicted labels for images with different occlusion rates, which are then compared to manually annotated ground truth labels. This evaluation measures correct detections, missed detections where ground truth objects are not identified, and false positives where non-existent objects are detected. The process involves reading both predicted and ground truth labels, converting them into a comparable format, matching them, and calculating the number of missed detections and false positives for each condition. The final accuracy is computed based on the total number of objects, providing insights into the model's performance under different levels of occlusion.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Setting of Experimental Environment and Training Parameters

The hardware used for this study's object detection model training includes an NVIDIA GeForce RTX 4060 GPU with 8GB VRAM and an Intel Core i7 processor. The software environment is based on Windows 11 Home Edition 64-bit with Python 3.8.5, using PyTorch 2.0.0 for building the model, and CUDA 11.8 for GPU acceleration. The model training was configured with 120 epochs to ensure convergence and optimize the performance under various occlusion conditions. A batch size of 4 was used due to hardware limitations, allowing for precise adjustments of model weights with each training step. Input images were resized to a resolution of 640x640 pixels, balancing accuracy and computational efficiency. The default YOLO optimizer (Adam or SGD) was utilized with standard learning rates and hyperparameters tuned for robustness in complex detection tasks.

B. Datasets

The dataset used in this study is based on Lin's dataset[7], consisting of 2,750 images: 1,800 synthetic generated by Data augmentation techniques such as rotation and translation likes Figure 5, 700 real-world, and 250 background images without objects. Synthetic images were designed to mimic occlusion scenarios, helping the model learn to detect objects under different angles and lighting. Real-world images were sourced from simulated industrial environments, capturing various

poses and stacking conditions of tools to improve adaptability. Background images help differentiate foreground objects from the background. The dataset includes five tool types: pliers (1,192), scissors (1,174), screwdrivers (1,193), glue sticks (1,174), and tape (1,127), providing a balanced and varied training set.



Figure 5. Fused images of different tools

C. Comparison of different basic models

Table I presents a comparison of several basic YOLO models YOLOv5l, YOLOv7, YOLOv8n, YOLOv8s in terms of performance under different levels of occlusion. The metrics include mAP@0.5, mAP@0.5:0.95, and accuracy under 20%, 40%, and 60% occlusion. The goal of this table is to assess the robustness and accuracy of each model in detecting objects, especially as the level of occlusion increases. This comparison aims to highlight the YOLOv8 models, particularly YOLOv8s, under challenging conditions involving occluded objects.

TABLE I. YOLO MODEL PERFORMANCE ACROSS OCCLUSION LEVELS

Model	mAP@.5	mAP@.5:.95	T20%	T60%
Yolov5l	0.992	0.803	99.7	80.6
Yolov7	0.996	0.787	99.2	87.6
Yolov8n	0.991	0.788	99.3	83.5
Yolov8s	0.993	0.811	99.5	88.8

YOLOv8s shows notable improvements in high occlusion scenarios. At 60% occlusion, it achieves 88.8% accuracy, surpassing both YOLOv5l at 80.6% and YOLOv7 at 87.6%, indicating stronger robustness in complex, high-occlusion scenes. Although YOLOv7 slightly leads in overall mAP@0.5 (99.59% vs 99.3% for YOLOv8s), YOLOv8s excels in detecting highly occluded objects. Despite being a lightweight model, YOLOv8n performs better than YOLOv7 in mAP@0.5:.95 (0.78754 vs 0.7866), showing a good balance between speed and accuracy. Comparing YOLOv8s to YOLOv8n, YOLOv8s consistently outperforms across all metrics, particularly in T60% occlusion (88.8% vs 83.5%) and achieves a higher mAP@0.5:.95 at 0.811. This highlights YOLOv8s's superior performance in more challenging object detection tasks. The subsequent comparison experiments will be improved based on the yolov8s model

D. Comparison of different improvement models based on yolov8s

By comparing the base YOLOv8s model with versions incorporating improvements and Lin's yolov5l with CA attention mechanism and X detect head, SIOU loss function

improvement model[7] with the same dataset this paper used, the purpose is to assess the impact of these enhancements on detection accuracy and robustness in occluded scenarios. Based on the experimental results across different occlusion levels T20%, T40%, T60%, the models performed as follows, where A(CBAM), B(AKConv), C(Effciou):

TABLE II. COMPARISON OF IMPROVED YOLOV8S MODELS

Model	mAP@.5	mAP@.5:.95	T20%	T60%
Yolov8s	0.993	0.811	99.5	88.8
Yolov8s+A	0.980	0.791	99.3	88.0
Yolov8s+A+B	0.990	0.789	99.2	91.2
Yolov8s+A+B+C	0.992	0.807	99.4	91.8
Lin-YoloModel	0.992	0.813	99.6	86.6

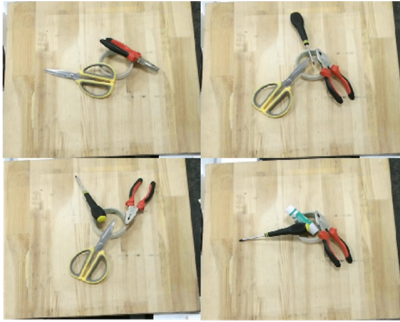


Figure 6. Images with 60% occlusion

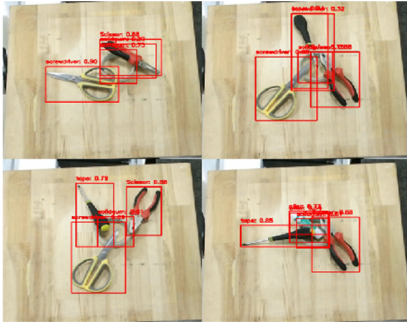


Figure 7. Visualization of YOLOv8s test result

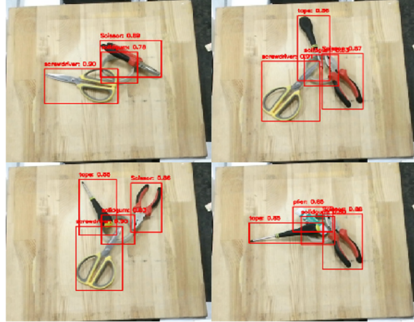


Figure 8. Visualization of CAE-YOLOv8 test results

The experimental results in Table II highlight significant improvements at T60%, with the "Yolov8s+A+B+C" model achieving an accuracy of 91.8%, surpassing the baseline Yolov8s (88.8%) and Lin-YOLOv5 (86.6%). This indicates the effectiveness of the proposed enhancements in handling high occlusion scenarios. For other metrics such as mAP@0.5 and lower occlusion levels T20%, the changes remain relatively small, suggesting that the improvements are particularly effective in challenging, heavily occluded environments.

Based on the comparison of Figures 6, 7, and 8, it is evident that the CAE-YOLOv8s model demonstrates a significant improvement in detection performance under high occlusion conditions (Zoom-in of the intercept at 60% occlusion). Compared to the original YOLOv8s, CAE-YOLOv8s is able to more accurately identify objects even when they are partially obscured, thereby enhancing detection precision and stability. These experimental results indicate that CAE-YOLOv8s has superior robustness and adaptability in handling occlusion challenges in complex scenes, resulting in a notable improvement in overall object detection performance.

IV. CONCLUSIONS

The goal of object detection in industrial settings is to boost automation efficiency, especially in challenging occluded scenarios. This study proposed an improved YOLOv8 model to address these issues. The CBAM attention mechanism was integrated after the SPPF layer to enhance focus on key features, while the AKConv module was added to increase adaptability to various object shapes. Additionally, the Effciou loss function replaced the standard IoU loss to improve bounding box accuracy, especially in heavy occlusions. The model achieved a 91.8% accuracy under 60% occlusion, showcasing its robustness in complex visual tasks.

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